EXP. NO: 3

DATE: 23/7/24

STUDY OF DIFFERENT TYPES OF NETWORK CABLES AND CRIMPING OF CABLE WITH RJ45 CONNECTOR

AIM:

To study different types of network cables and perform crimping of a UTP cable using an RJ45 Connector

MATERIALS REQUIRED:

- UTP (cat 5e / cat6) cable
- RJ45 Connector
- Crimping tool
- Wire stripper
- Cable cutter
- LAN cable tester

TYPES OF NETWORK CABLES:

=> UTP (unshielded Twisted Pair): Most Commonly used for Ethernet Networks.

Speed : upto 40Gbps
Range : 100meters

=> STP (shielded Twisted Pair): Has shielding to reduce electromagnetic interference

Speed : upto 40Gbps
range : 100meters

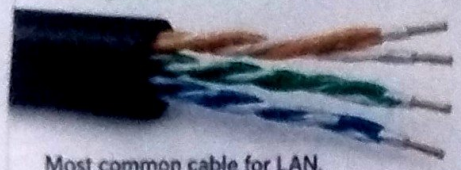
=> Coaxial cable : used for cable Tv and older Networking systems.

Speed : upto 10Gbps
range : 200 to 500 meters

TYPES OF NETWORK CABLES

Choose the Right Cable for Reliable & High-Performance Networks

1 UNSHIELDED TWISTED PAIR (UTP) CABLES



Most common cable for LAN, Ethernet, and office networks.

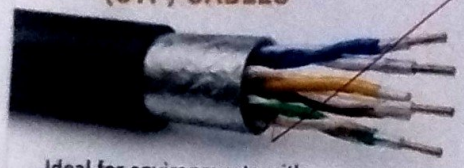
STRUCTURE

- Outer Jacket
- Twisted Pairs

KEY BENEFITS

- Commonly used in homes and offices
- Flexible, affordable, and easy to use
- For Ethernet and VoIP applications

2 SHIELDED TWISTED PAIR (STP) CABLES



Ideal for environments with high EMI and interference.

STRUCTURE

- Outer Jacket
- Foil Shielding
- Twisted Pairs

KEY BENEFITS

- Protect against interference and crosstalk
- Require proper grounding
- Ideal for environments with heavy EMI such as factories and industrial setups

3 COAXIAL CABLES



Used for traditional broadband, CCTV, and RF transmissions.

STRUCTURE

- Outer Protective Jacket
- Metallic Shield
- Copper Conductor

KEY BENEFITS

- Highly resistant to signal obstruction
- Great for long-distance transmissions
- Commonly used in data centers, ISP networks, and radio/broadcast applications

4 FIBER OPTIC CABLES



Built for speed and performance in modern networks.

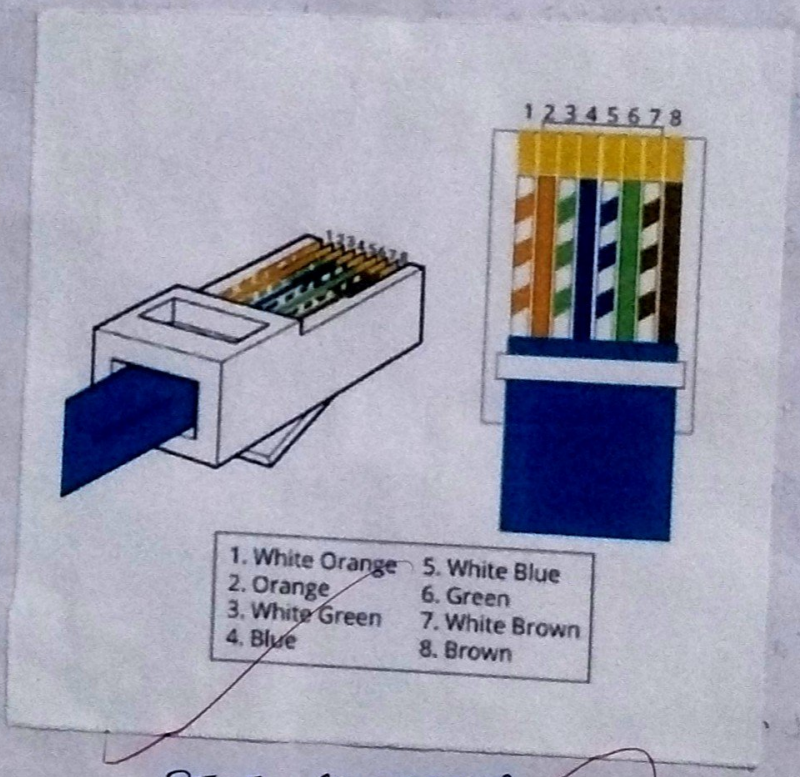
STRUCTURE

- Outer Jacket
- Glass Fiber Strands
- Light Signal Transmission

KEY BENEFITS

- Immune to electromagnetic interference
- Ideal for ultra-fast, long-range data transmission
- Widely used in data centers, ISP networks, and high-bandwidth applications

Tip: Selecting the right cable improves network performance, reduces downtime, and ensures a reliable network infrastructure.



RJ45 CONNECTOR

=> Fiber optic cable: uses light to transmit data at very high speeds over long distances

Speed: 100+ Gbps Range: upto 40 Kilometers

CRIMPING PROCEDURE:

* Cut the cable to the required length
* Strip about 2-3cm of the outer insulation
* Untwist and arrange the wires according to the T568B standard:
  - White/orange
  - orange
  - White/Green
  - Blue
  - White/Blue
  - Green
  - White/Brown
  - Brown
* Trim the wires evenly
* Insert the wires fully into the RJ45 connector
* Use the crimping tool to crimp the connector
* Test the cable using a LAN cable tester.

RESULT:

Thus the network cable was successfully crimped with an RJ45 connector and tested for proper connectivity